# Experiment 4

## Title

Develop a dynamic webpage where content updates on button click using JavaScript DOM. Implement jQuery effects: hide/show elements, toggle visibility, and animate elements.

## Theory:

1. **JavaScript and the DOM (Document Object Model):**
    - The **DOM** is a hierarchical representation of the elements on a webpage. It allows programming languages like JavaScript to interact with the structure, style, and content of the page.
    - Using **DOM methods**, we can:
        - Access elements (getElementById, getElementsByClassName, querySelector)
        - Change content (innerHTML, textContent)
        - Modify attributes (setAttribute, removeAttribute)
        - Change styles dynamically (style.property)
    - Example: Changing the text of a paragraph when a button is clicked enhances user interactivity.
2. **Dynamic Webpage Content:**
    - A **dynamic webpage** changes content without reloading the page.
    - Dynamic updates improve **user experience** by allowing real-time feedback and interactivity.
    - In this lab, content updates are achieved using document.getElementById().innerHTML to manipulate text dynamically.
    -
3. **Introduction to jQuery:**

- **jQuery** is a lightweight, fast JavaScript library that simplifies:
  - HTML document traversal and manipulation
  - Event handling
  - Animations and effects
- jQuery is **cross-browser compatible**, meaning the same code works in different browsers without modification.

Syntax Example:

```
$(selector).action();
```

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  - $() is the jQuery function
  - selector identifies the HTML element
  - action() is the method applied (like hide(), show(), animate())

4. **jQuery Effects and Animations:**
   - **Hide/Show:** Instantly hide or show elements.
   - **Toggle:** Switch visibility state of an element (hide → show, show → hide).
   - **Animate:** Allows smooth transitions of CSS properties (like height, width, opacity, margins).
   - **Benefits of jQuery Effects:**
     - Improves visual interactivity
     - Reduces manual CSS manipulation
     - Provides smoother user experience
5. **Integration of JS DOM and jQuery:**

- Using **pure JavaScript** allows control over element content and structure.
- Using **jQuery** complements JS by handling animations and effects more efficiently.
- Combining both allows for **dynamic and interactive web pages** where content changes and user interactions are smooth and visually appealing.

6. **Applications:**
   - Dynamic webpages are used in:
     - Online forms that validate inputs instantly
     - Content sliders and carousels
     - Real-time notifications
     - Interactive dashboards

## Procedure

1. Include jQuery CDN:

```
<script src="https://code.jquery.com/jquery-3.7.0.min.js"></script>
```

2. Create HTML element:

```
<div id="box">Hello</div>
<button id="btn">Toggle</button>
```

3. Apply jQuery:

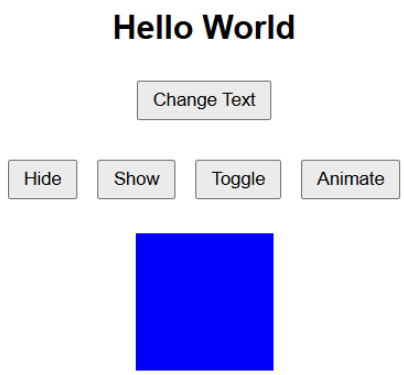
```
<script>
$('#btn').click(function(){
  $('#box').toggle(500);
});
</script>
```

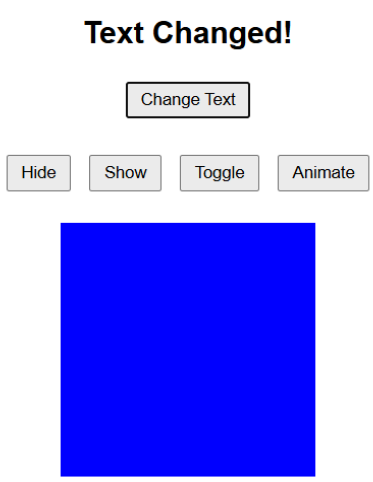
## Code:-

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>DOM & jQuery Effects</title>

  <!-- jQuery CDN -->
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 50px;
    }

    #box {
      width: 100px;
      height: 100px;
      background: blue;
      margin: 20px auto;
    }

    button {
      margin: 5px;
      padding: 5px 10px;
    }
  </style>
</head>
<body>

  <h2 id="text">Hello World</h2>

  <!-- DOM update -->
  <button onclick="changeText()">Change Text</button>

  <br><br>
```

```
  <!-- jQuery effects -->
  <button id="hideBtn">Hide</button>
  <button id="showBtn">Show</button>
  <button id="toggleBtn">Toggle</button>
  <button id="animateBtn">Animate</button>

  <div id="box"></div>

  <script>
    // JavaScript DOM
    function changeText() {
      document.getElementById("text").innerText = "Text Changed!";
    }

    // jQuery Effects
    $("#hideBtn").click(function () {
      $("#box").hide();
    });

    $("#showBtn").click(function () {
      $("#box").show();
    });

    $("#toggleBtn").click(function () {
      $("#box").toggle();
    });

    $("#animateBtn").click(function () {
      $("#box").animate({
        width: "200px",
        height: "200px"
      });
    });
  </script>

</body>
</html>
```

# Output:-


Hello World
Change Text
Hide
Show
Toggle
Animate


Text Changed!
Change Text
Hide
Show
Toggle
Animate

## Conclusion

JavaScript DOM allows dynamic updating of webpage content in real-time. jQuery simplifies element manipulation, effects, and animations. Combining both creates interactive and visually appealing webpages, enhancing user experience. This lab demonstrated practical implementation of content updates, visibility toggling, and animations.

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## Question:

1. What is the difference between .hide() and .toggle() in jQuery?
   `.hide()` hides the selected element.
   `.toggle()` switches the element between hidden and visible each time it is called.

2. How can you **stop an ongoing animation** in jQuery before it completes?
   Using the `.stop()` method.

3. How can you use jQuery to **fade in** and **fade out** an element?

4. Use `.fadeIn()` to show the element gradually.

5. Use `.fadeOut()` to hide the element gradually.

Example:

```
$("#box").fadeIn();

$("#box").fadeOut();
```